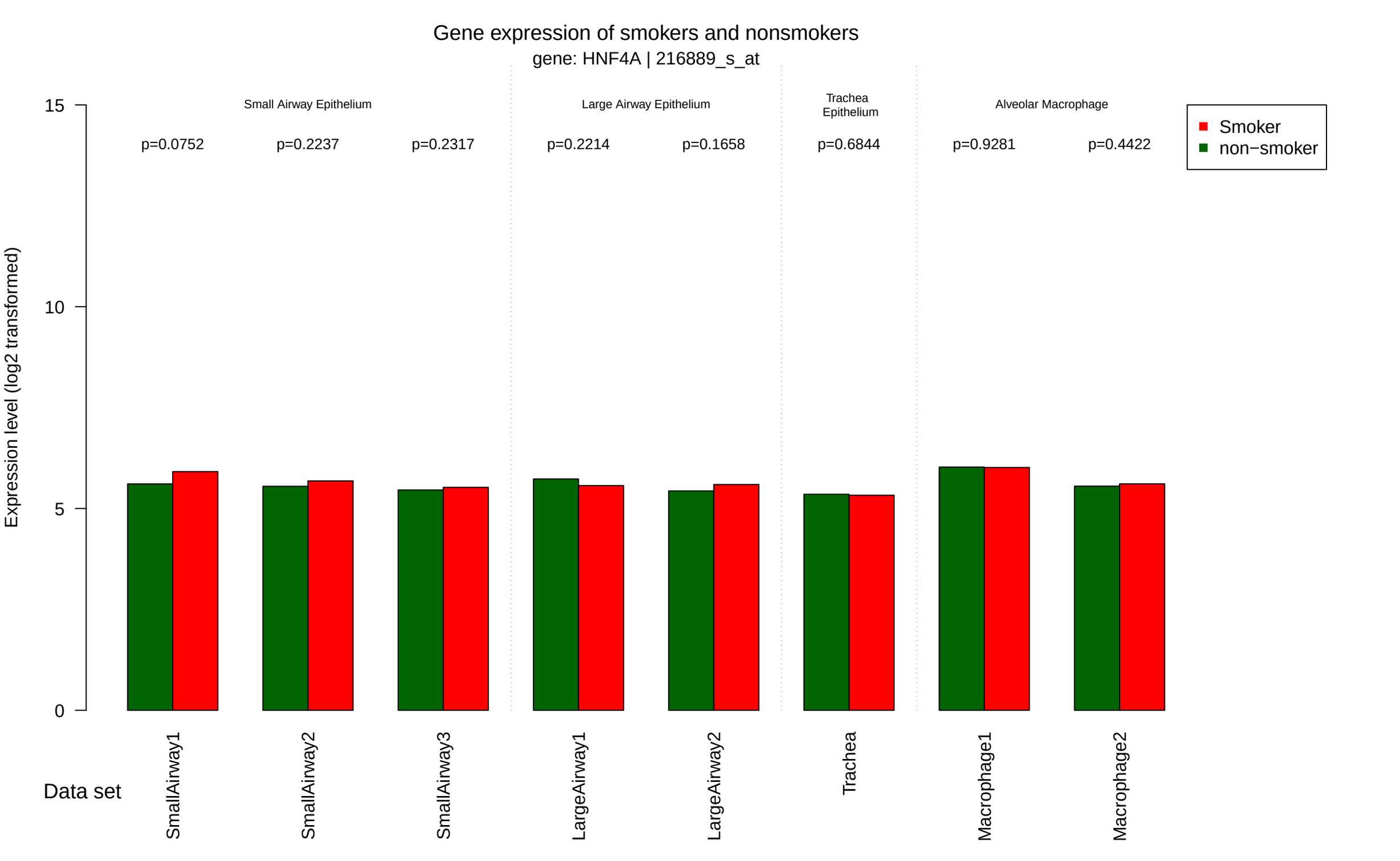
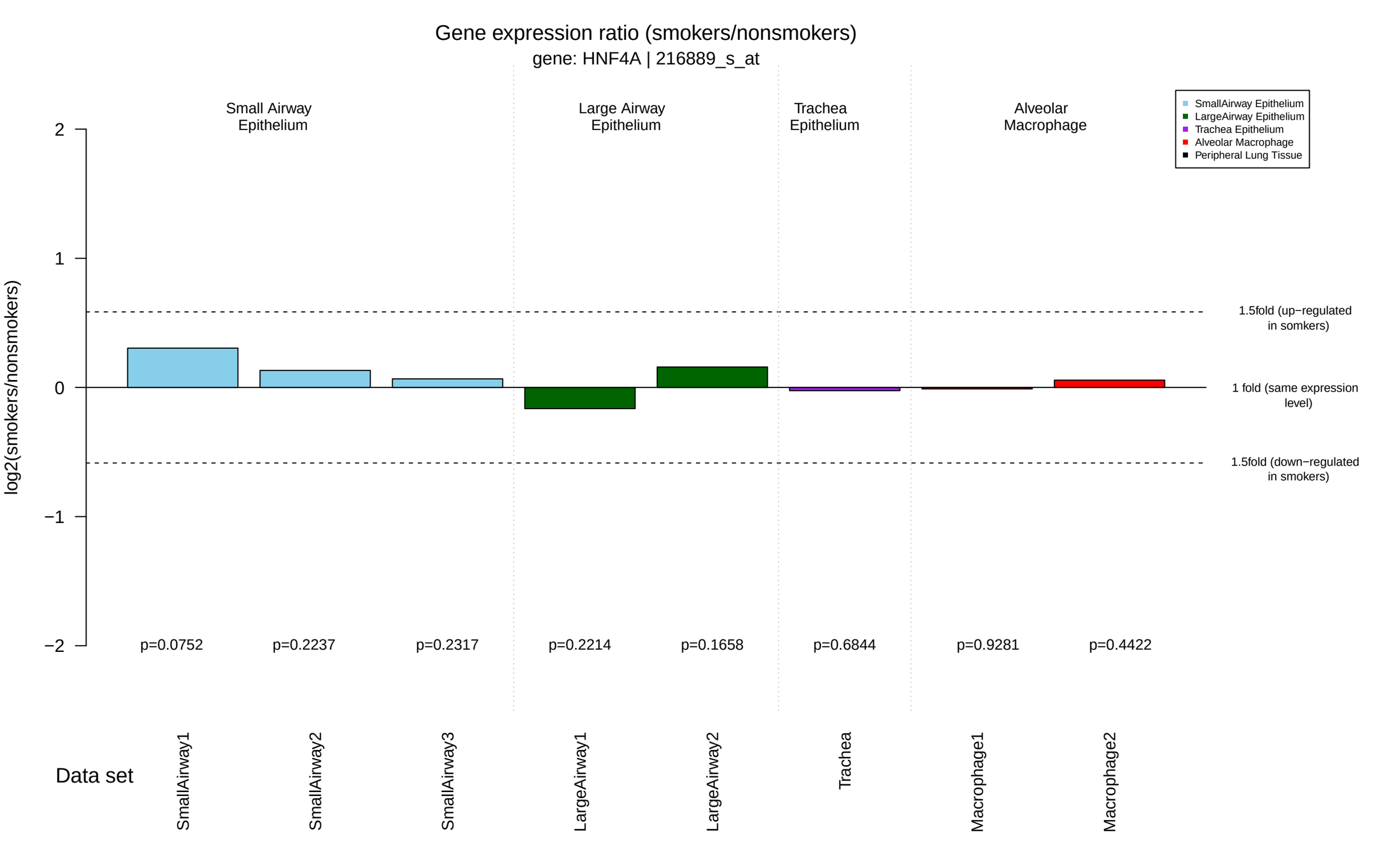
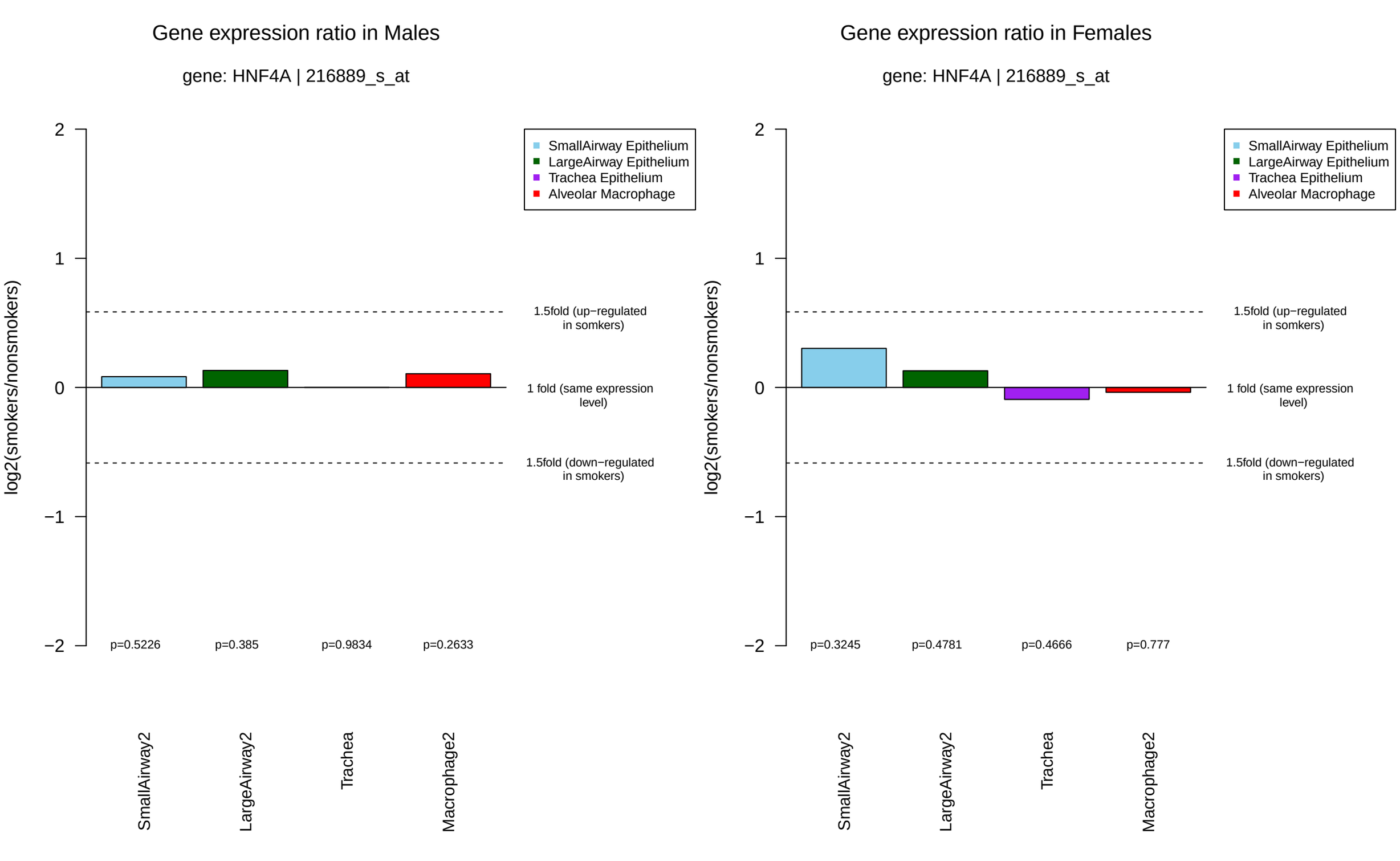
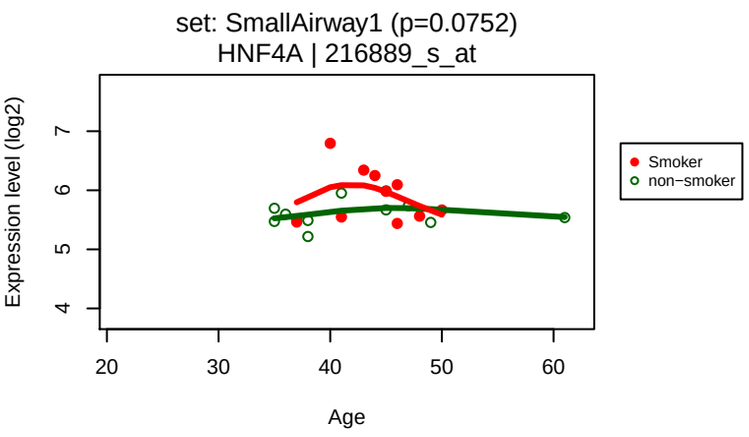
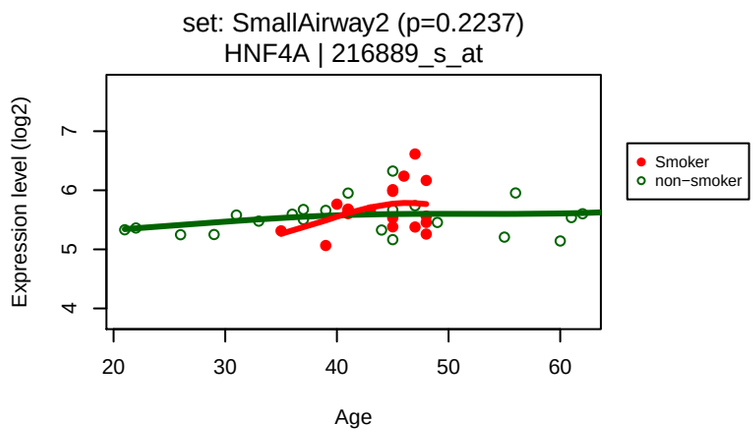
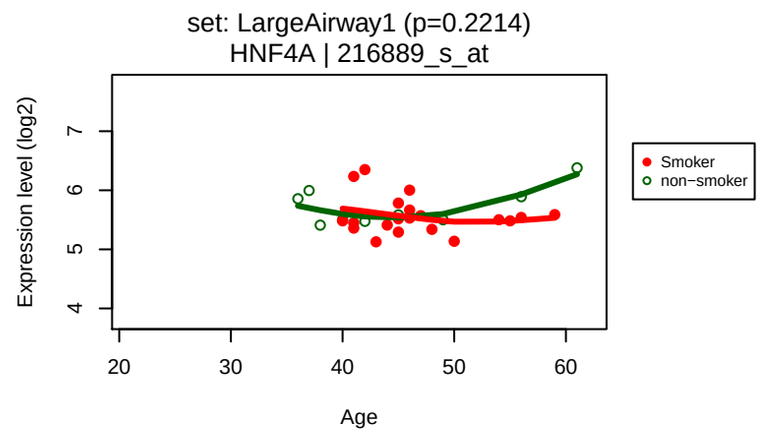

Gene expression of smokers and nonsmokers
gene: HNF4A | 216889_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0752
p=0.2237
p=0.2317
p=0.2214
p=0.1658
p=0.6844
p=0.9281
p=0.4422
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

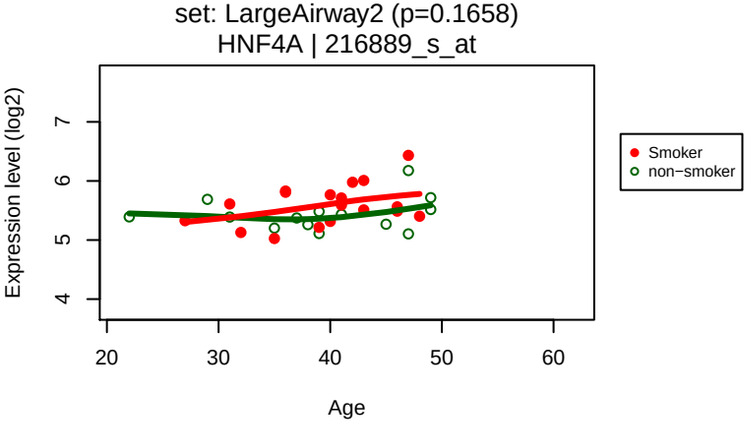

Gene expression ratio (smokers/nonsmokers)
gene: HNF4A | 216889_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0752
p=0.2237
p=0.2317
p=0.2214
p=0.1658
p=0.6844
p=0.9281
p=0.4422
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

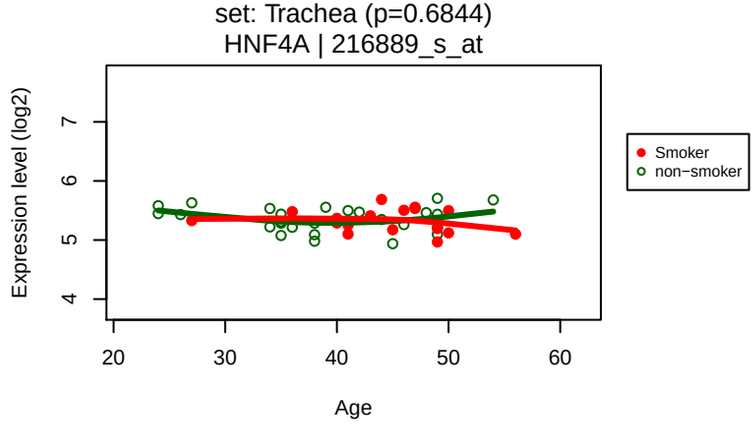

Gene expression ratio in Males
gene: HNF4A | 216889_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5226
p=0.385
p=0.9834
p=0.2633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HNF4A | 216889_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3245
p=0.4781
p=0.4666
p=0.777
SmallAirway2
LargeAirway2
Trachea
Macrophage2

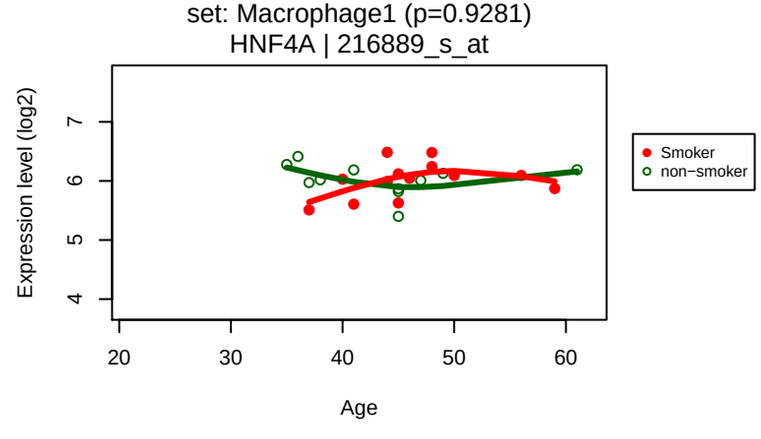

set: SmallAirway1 (p=0.0752)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker

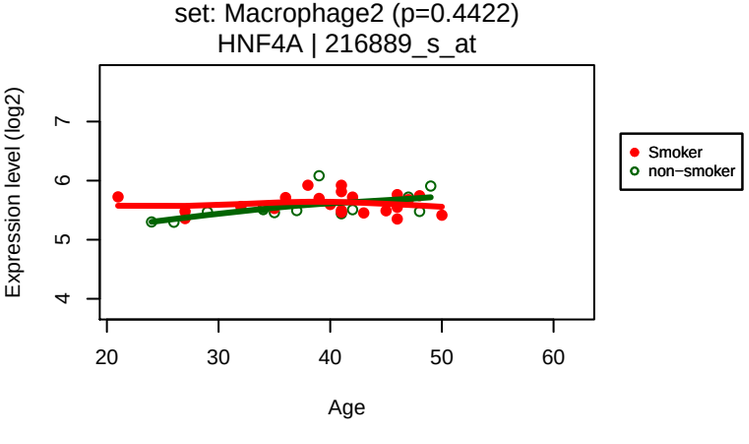

set: SmallAirway2 (p=0.2237)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker

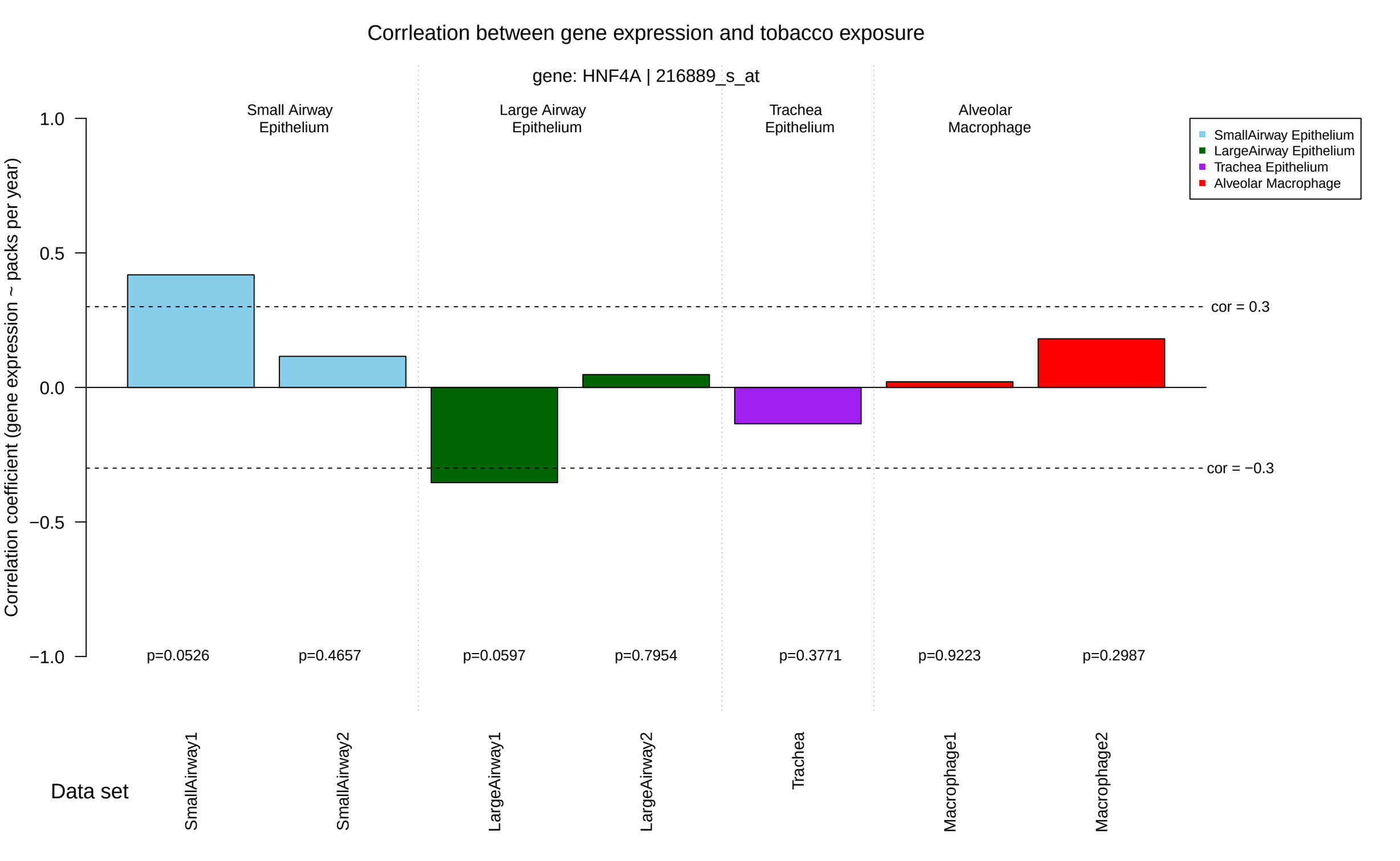

set: LargeAirway1 (p=0.2214)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1658)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6844)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9281)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4422)
HNF4A | 216889_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HNF4A | 216889_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0526
p=0.4657
p=0.0597
p=0.7954
p=0.3771
p=0.9223
p=0.2987
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2